DIAGNOBOT

Executive Summary

DiagnoBot is a student-built prototype for image-assisted health triage. The software combines a fine-tuned MobileNetV3-Small classifier for ten skin-image folder labels, Moondream via Ollama for image descriptions, Llama 3.2 via Ollama for short text summaries, and deterministic Python rules for symptom extraction and urgency cues. The broader codebase also contains a TorchXRayVision DenseNet-121 chest X-ray model path. The latest skin model was trained on 932 images and achieved 70.7% accuracy, 70.3% balanced accuracy, and 0.699 macro-F1 on a 140-image internal test split. The test split is image-level, not patient-level, and there is no external test cohort; these results are preliminary and do not establish clinical performance.

The physical concept uses an available Raspberry Pi 5, ESP32, L298N, chassis, motors, and planned MAX30102, MLX90614, and USB webcam. The repository does not yet implement Pi GPIO, ESP32 serial, sensor reads, or motor control. This report separates working software from the proposed hardware build and provides recent paper references for the literature review.

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1. Project Overview

The project aims to demonstrate how image analysis, basic physiological sensing, local AI models, and a human-facing interface could be combined in a low-cost health-triage robot. The current repository is primarily a FastAPI web application with dashboard pages and ML pipeline modules. It has a locally stored ten-class skin checkpoint, a separate X-ray route, local vision/text model paths, and a deterministic triage layer.

1.1 Current software modules

Module	Current role
Skin image classifier	MobileNetV3-Small returns one of ten dataset labels and prediction confidence.
Vision-language model	Moondream describes image content; it is not the skin classifier.
Clinical text model	Llama 3.2 summarizes supplied context; no answer-accuracy benchmark exists.
Chest X-ray path	TorchXRayVision DenseNet-121 with MIMIC-Ch weights; not evaluated in this report.
Rules	Python extraction and urgency cues; deterministic, not learned.
Web service	FastAPI endpoints and dashboards for image uploads, reports, and triage.

The app defaults to MOCK_MODE=true, so model-backed calls may be replaced by simulation/fallback behavior. When enabled, the LLM/VLM use Ollama tags `llama3.2` and `moondream`. These are tags, not pinned versions; record model digests for reproducibility.

2. Problem Statement and Objectives

2.1 Problem statement

The system is intended to collect a skin image and basic vital signs, process them locally, and present an understandable summary for human review. The current software and physical prototype are not fully integrated: the dataset provenance and labels are not adequately documented, the MAX30102/MLX90614/webcam are planned purchases, and the code repository has no Pi/ESP32 sensor or motor-control implementation. The classifier has only 140 internal test images and no patient/source identifiers, so related images could occur across partitions.

2.2 Project objectives

- Provide a local user interface and API for skin-image input and decision-support output.
- Train and evaluate a compact ten-class MobileNetV3-Small skin classifier.
- Describe separate roles for the skin classifier, local LLM/VLMs, and deterministic triage rules.
- Plan sensor, camera, display, and chassis integration using the components available to the team.
- Report measurable results and state limits before claiming medical usefulness.

2.3 Scope

Included: software architecture, skin-model training/evaluation, local model roles, hardware plan, and student-prototype demonstration. Not yet demonstrated: clinical diagnosis, external validation, live sensor acquisition, Pi inference performance, motor control, or end-to-end robot testing.

3. Literature Review - Recent Papers

The selected 2024-2025 studies cover data efficiency, label noise, fairness, model fusion, clinical data documentation, embedded sensing, and reporting standards. Their numerical results use different datasets and test protocols; they are context, not direct competitors to DiagnoBot.

[1] Chen, Liu, and Yang (2024) - CDD-Net and Derm104. This few-shot classification study introduced 4,483 clinical images across 104 categories and a multiscale feature-fusion module. It reported gains up to 9.14 percentage points for selected few-shot settings. Relevance: small labeled datasets are common in dermatology; however, category-holdout few-shot evaluation differs from DiagnoBot's ordinary ten-class image split.

[2] Shen et al. (2024) - learning from online community images. The authors used contrastive pretraining and label filtering for 22 skin diseases. Reported top-1 accuracy increased from 42.05% to 45.05% after pretraining and reached 49.64% after filtering with a larger validation set. Relevance: correcting noisy labels can matter as much as architecture changes. Internet images can introduce source and population bias.

[3] Dip et al. (2024) - equitable skin prediction. This transfer-learning and domain-adaptation study evaluated Diverse Dermatology Images, which includes underrepresented and common skin tones. It highlights that average accuracy can mask subgroup performance gaps. DiagnoBot has no skin-tone metadata or subgroup evaluation, so representative data and subgroup checks are future requirements.

[4] Alruwaili and Mohamed (2025) - multi-architecture classification. The study combined EfficientNet-B0, EfficientNet-B2, and ResNet50 on 27,153 Kaggle images and reported 99.14% accuracy using an 80/10/10 split. The result is not directly comparable to DiagnoBot: data scale, labels, and split differ, and image-level random splits may inflate results when related images overlap. It illustrates why accuracy must be interpreted with dataset and split details.

[5] Yan et al. (2025) - PanDerm. This dermatology foundation model used over two million multimodal images from 11 institutions and four imaging modalities, with broad benchmark and reader studies. It is far larger than a student model and is not a direct comparator. Relevance: diverse datasets, multiple modalities, and clinician-reader evaluation strengthen medical AI evidence.

[6] Sultana et al. (2025) - SkinDisNet dataset. The paper describes 1,710 smartphone images from hospitals in Bangladesh across six conditions, with clinical/demographic metadata. Labels differ from DiagnoBot, but its provenance and metadata documentation demonstrate the value of traceable data.

[7] Pawar et al. (2024) - ESP32 health-monitoring prototype. This conference prototype combines an ESP32, MAX30102, temperature sensor, display, and connected app. It is relevant to the embedded-sensing plan; it does not validate DiagnoBot or demonstrate clinical-grade readings.

[8] Tejani et al. (2024) - CLAIM 2024 reporting guidance. The checklist recommends transparent reporting of data source, partition sizes and split unit, preprocessing, evaluation, and external testing. It distinguishes internal held-out testing from a separate external dataset and generally recommends disjoint patient-level or higher partitions.

3.1 Main lessons for DiagnoBot

- Improve label quality and document image provenance before changing the model architecture.
- Use patient/source-grouped splits and independent external testing when possible.

- Measure performance by class and relevant subgroup; aggregate accuracy alone is insufficient.
- Evaluate sensors, generated language, rules, and the image classifier as separate subsystems.

4. Proposed Solution and Model Selection

The proposed solution combines the current local application with a planned robot front end. The camera supplies images; the skin classifier provides class probabilities; a vision-language model generates a separate observation; a language model summarizes supplied context; deterministic rules extract warning signals; and the display presents each output for human review. No generated response should be presented as a confirmed diagnosis.

4.1 Exact model inventory and rationale

Component	Model used in code	Why / intended role
Skin classification	Torchvision MobileNetV3-Small, ImageNet default weights, fine-tuned in PyTorch	Compact transfer-learning model for CPU/edge experiments; ten outputs correspond to local folder names.
Image description	Moondream via local Ollama tag `moondream`	Generates text observations about the image; separate from classification and not evaluated here.
Clinical summary	Llama 3.2 via local Ollama tag `llama3.2`	Generates concise context summaries; no project answer-quality metric is implemented.
Chest X-ray	TorchXRayVision DenseNet-121, `densenet121-res224-mimic_ch`	Separate pretrained chest X-ray path in the larger codebase; outside skin-model evaluation.
Extraction/triage	Project Python rules	Deterministic and inspectable; not an LLM, but not clinically validated.

4.2 Proposed architecture

The software workflow and proposed physical robot connections are shown separately for clarity.

Software workflow

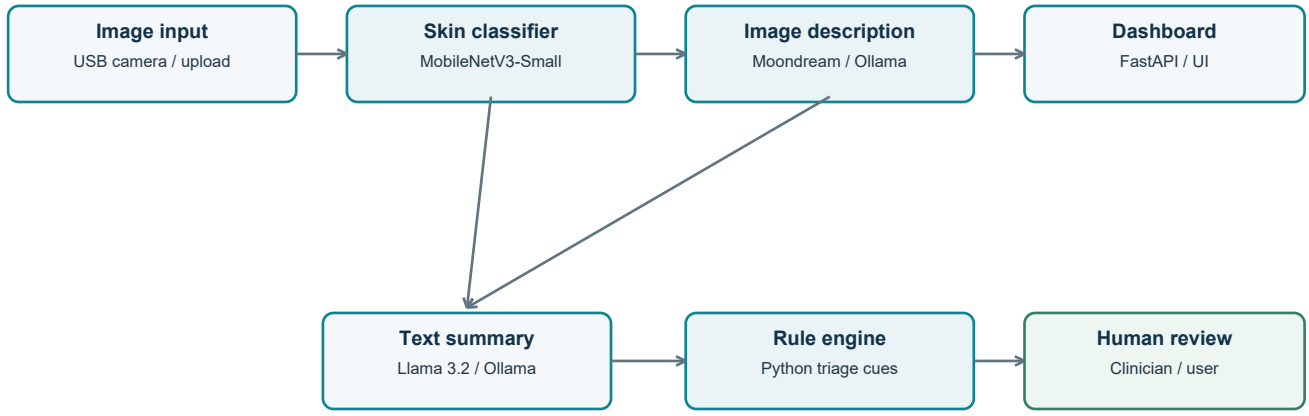


Figure 1. Current software path: image input, classifier, language-model paths, rules, dashboard, and human review.

PLANNED HARDWARE LINKS — NOT YET IMPLEMENTED

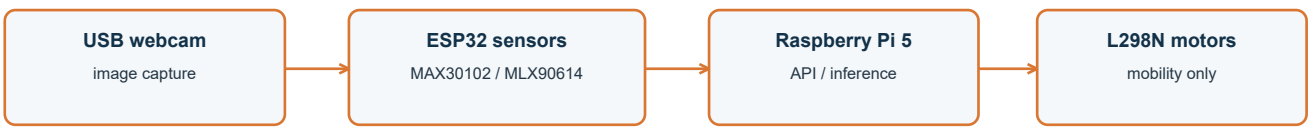


Figure 2. Planned hardware links; sensor acquisition and motor-control software are not yet implemented.

5. Working Principle

Software workflow: (1) user uploads or captures an image; (2) the image is resized to 224 x 224 and passed to MobileNetV3-Small; (3) class scores and confidence are formatted for the dashboard; (4) Moondream can provide a separate image description; (5) symptom/vital information is parsed by deterministic rules; (6) Llama 3.2 can summarize the supplied context; and (7) the user or clinician reviews the findings. The LLM output does not replace classifier evidence or a clinician.

Planned robot: ESP32 reads MAX30102 and MLX90614 over I2C/SMBus and sends timestamped values by serial to the Pi. The Pi runs the service and reads the USB webcam. L298N drives motors from a separate supply. Stop motors for readings.

6. Methodology

6.1 Dataset and split

The local folder-organized dataset contains 932 images and ten labels. A seeded (42), stratified image-level split assigned 652 training images, 140 tuning/validation images, and 140 internal test images. Patient/source IDs were unavailable. Dataset origin, license, clinical reference standard, annotator qualifications, and label adjudication are not documented in the repository and must be verified before formal dissemination.

Table 1. Image counts by class

Class	Train	Tune	Test
Bullous	70	15	15
Eczema Photos	70	15	15
Lupus	70	15	15
Urticaria Hives	70	15	15
acanthosis nigricans	44	9	10
acne	70	15	15
candidiasis	70	15	15
impetigo	50	11	11
molluscum-contagiosum	68	15	14
nevus	70	15	15
Total	652	140	140

6.2 Preprocessing and training

Images were resized to 224 x 224 pixels, converted to tensors, and normalized with ImageNet channel statistics. Training augmentation used horizontal flips (p=0.5), rotations up to 10 degrees, and mild brightness/contrast/saturation jitter (0.12). Training transforms were kept separate from tuning and test transforms. Batch size was 32.

MobileNetV3-Small used ImageNet default pretrained weights and a replaced final layer with ten outputs. The loss was class-weighted cross-entropy, with weights proportional to the square root of inverse training frequency. The classifier head was trained up to five epochs with AdamW at 5e-4 learning rate and 1e-4 weight decay. The final four feature blocks were then fine-tuned for up to twelve more epochs at 2e-5 learning rate and 1e-4 weight decay. The run reached the 17-epoch cap; the best tuning macro-F1 checkpoint was epoch 15. The held-out test set was evaluated once after checkpoint selection.

Training ran on CPU for approximately 351 seconds. At epoch 15, training accuracy was 81.75% and tuning accuracy 67.14%, a gap of about 14.6 percentage points. This indicates possible overfitting, but it is not a reliable estimate of future performance because the dataset and test split are small.

7. Prototype and Hardware Plan

Reported available parts are Raspberry Pi 5, ESP32, L298N motor driver, chassis, DC motors/wheels, breadboard, and jumper wires. Planned purchases are an MLX90614/GY-906 temperature module, MAX30102 heart-rate/SpO2 module, and USB webcam. The supplied photograph shows a Raspberry Pi connected to a display running the DiagnoBot interface.

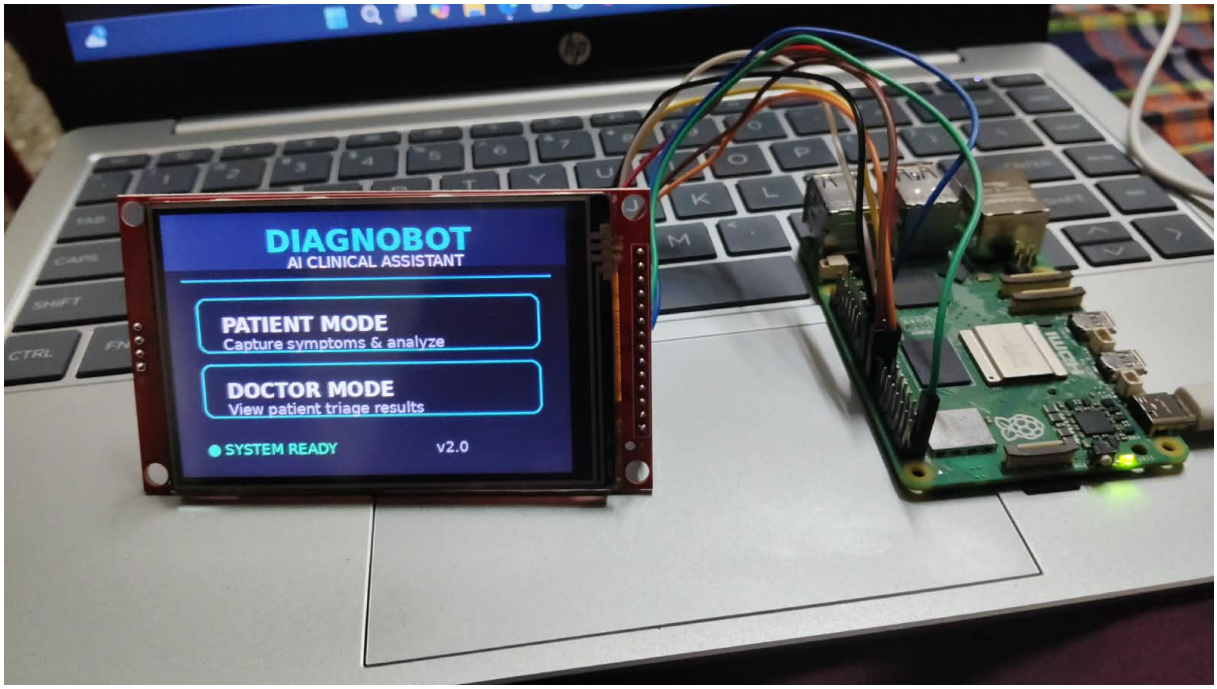


Figure 3. User-provided bench photograph of the Pi and DiagnoBot display. It does not demonstrate sensor reads, motor control, or full integration.

Table 2. Current implementation status

Feature	Evidence/status
Skin classifier	Implemented; internal image evaluation is available.
FastAPI and dashboard	Present in the repository.
Moondream and Llama 3.2	Ollama call paths exist when mock mode is disabled; text quality is not benchmarked.
Chest X-ray model	DenseNet-121 loading path exists; not evaluated in this report.
Pi GPIO / ESP32 serial / sensors	No controller or sensor code in repository; modules were listed as planned purchases.
Motor control	Hardware is listed but no motor-control code was found in repository.
End-to-end robot	Not evaluated.

Hardware safety notes

MAX30102 is an optical heart-rate/pulse-oximeter sensor; breakout boards differ in regulator and logic-level design, so check the exact board before wiring. MLX90614 is a non-contact infrared surface/object thermometer; it is not a direct core-body measurement. Raspberry Pi GPIO is 3.3 V logic. Stop motors during readings, isolate motor supply current from sensor rails, verify L298N limits, and use official component documentation [9-12].

8. Evaluation Metrics and Results

8.1 Metric definitions

Accuracy is correct predictions divided by all test images. Per-class precision = TP/(TP+FP), recall = TP/(TP+FN), and F1 = 2 x precision x recall / (precision + recall). Macro-F1 is the unweighted mean of class F1. Balanced accuracy is the mean recall across classes. Cross-entropy loss penalizes low probability assigned to the correct class. The confusion matrix records true versus predicted labels.

These metrics measure the skin image classifier only. They do not assess LLM answer quality, triage rules, sensor measurement accuracy, or whole-robot performance.

8.2 Per-class test results

Class	n	Precision	Recall	F1
Bullous	15	0.700	0.467	0.560
Eczema Photos	15	0.467	0.467	0.467
Lupus	15	0.889	0.533	0.667
Urticaria Hives	15	0.588	0.667	0.625
acanthosis nigricans	10	0.875	0.700	0.778
acne	15	0.700	0.933	0.800
candidiasis	15	0.938	1.000	0.968
impetigo	11	0.600	0.545	0.571
molluscum-contagiosum	14	0.647	0.786	0.710
nevus	15	0.778	0.933	0.848

Table 3. Class-level results on the held-out internal test set; n is the support count.

8.3 Overall results

Metric	Result	Meaning
Accuracy	70.7% (99/140)	Fraction of test images classified correctly.
Balanced accuracy	70.3%	Average recall across ten classes.
Macro precision	71.8%	Unweighted average of class precision.
Macro recall	70.3%	Unweighted average of class recall.
Macro-F1	0.699	Unweighted average class F1.
Test cross-entropy	0.915	Probability-sensitive loss.
Approx. 95% Wilson interval	62.8%-77.7%	Accuracy uncertainty for n=140; no correction for leakage or bias.

The approximate Wilson interval is based on 99 correct predictions among 140 images. It does not account for repeated patients, possible source leakage, label errors, or changes in camera/population. The lowest class F1 values were Eczema Photos (0.467), Bullous (0.560), and impetigo (0.571). Six of 15 Eczema Photos images were predicted as Urticaria Hives. The high Candidiasis F1 (0.968) is based on only 15 images and remains uncertain.

For a college project report, these figures are suitable as preliminary internal results when the split and limitations are stated. They are not clinical accuracy and are not directly comparable to papers with other datasets or split protocols.

9. Limitations and Recommendations

- Document dataset source, license, image acquisition, label definitions, and qualified reference-label process; detect exact and near-duplicate images.
- Use patient/source-grouped splits and reserve an untouched external cohort. If identifiers are missing, use a better-documented dataset or collect new governed data.

- Focus on one coherent research task with reliable labels, visually relevant mimics, and negative examples; add independent images across skin tones, phones, lighting, and sites.
- Compare a majority-class baseline and an alternate pretrained model using identical grouped splits; tune on training/tuning data only.
- Report per-class sensitivity/specificity, precision, macro-F1, balanced accuracy, confidence intervals, and calibration. Keep LLM, rules, and sensor evaluation separate.
- Validate LLM outputs with blinded human review and a prespecified rubric; compare sensor readings with appropriate reference instruments.
- Integrate hardware in stages and measure serial/I2C reliability, latency, power, sensor repeatability, and usability before a live demonstration.

10. Conclusion

DiagnoBot demonstrates a local software prototype that combines a compact skin image classifier, local language/vision model paths, deterministic triage logic, and a web interface. The retrained MobileNetV3-Small achieved 70.7% accuracy, 70.3% balanced accuracy, and 0.699 macro-F1 on a 140-image internal image-level test split. The result is useful for a student demonstration but does not establish clinical reliability. The small test, class confusion, undocumented dataset provenance, possible image leakage, and absence of external testing are important limitations.

The Pi 5, ESP32, sensor modules, webcam, and motor chassis form a feasible proposed hardware plan, but their software integration is not present in the repository. Next work should prioritize documented labels and grouped evaluation, one focused task, external testing, Pi performance measurements, and staged sensor integration. All output should remain subject to human review and should not be presented as a diagnosis.

References

A. Recent papers and reporting guidance

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B. Official component and platform references

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train_skin_mobilenet.py; ml_pipeline/vision/derm_cnn.py; ml_pipeline/vision/qwen_vl_analyzer.py; ml_pipeline/nlp/report_generator.py; ml_pipeline/vision/xray_analyzer.py; config/settings.py; evaluation/skin_test_metrics.json; evaluation/skin_test_confusion_matrix.csv. These are project evidence, not published papers.

Before college submission: fill in the author, guide, and institution fields and confirm the dataset source/permissions and local model versions.